THE CIRCLE

CONICAL SECTIONS

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

40%

WEEKLY

60%

EXAM

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

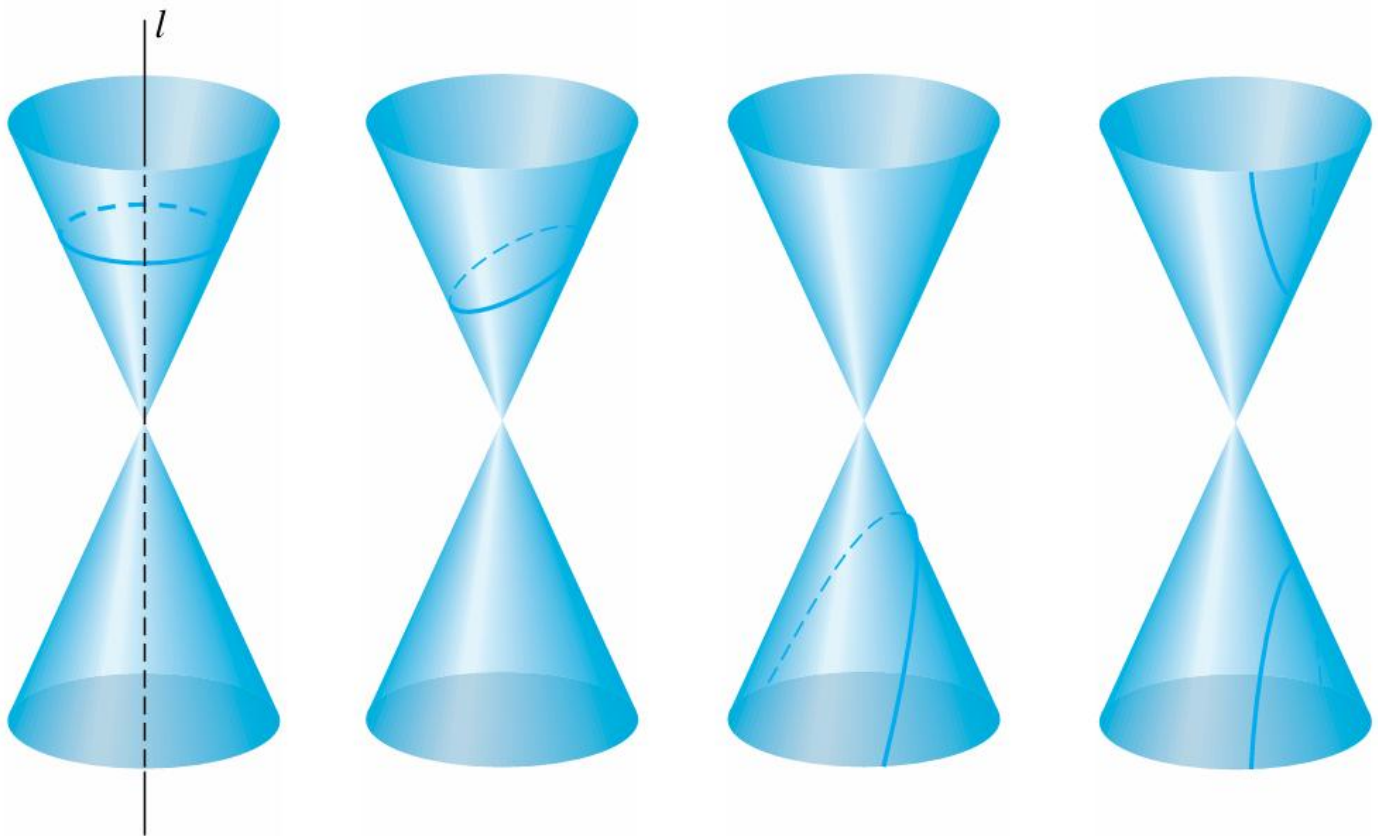
TOPIC OUTLINE

Circle with Center at the Origin

Circle with Center at $C(h, k)$

CIRCLE WITH CENTER AT THE ORIGIN

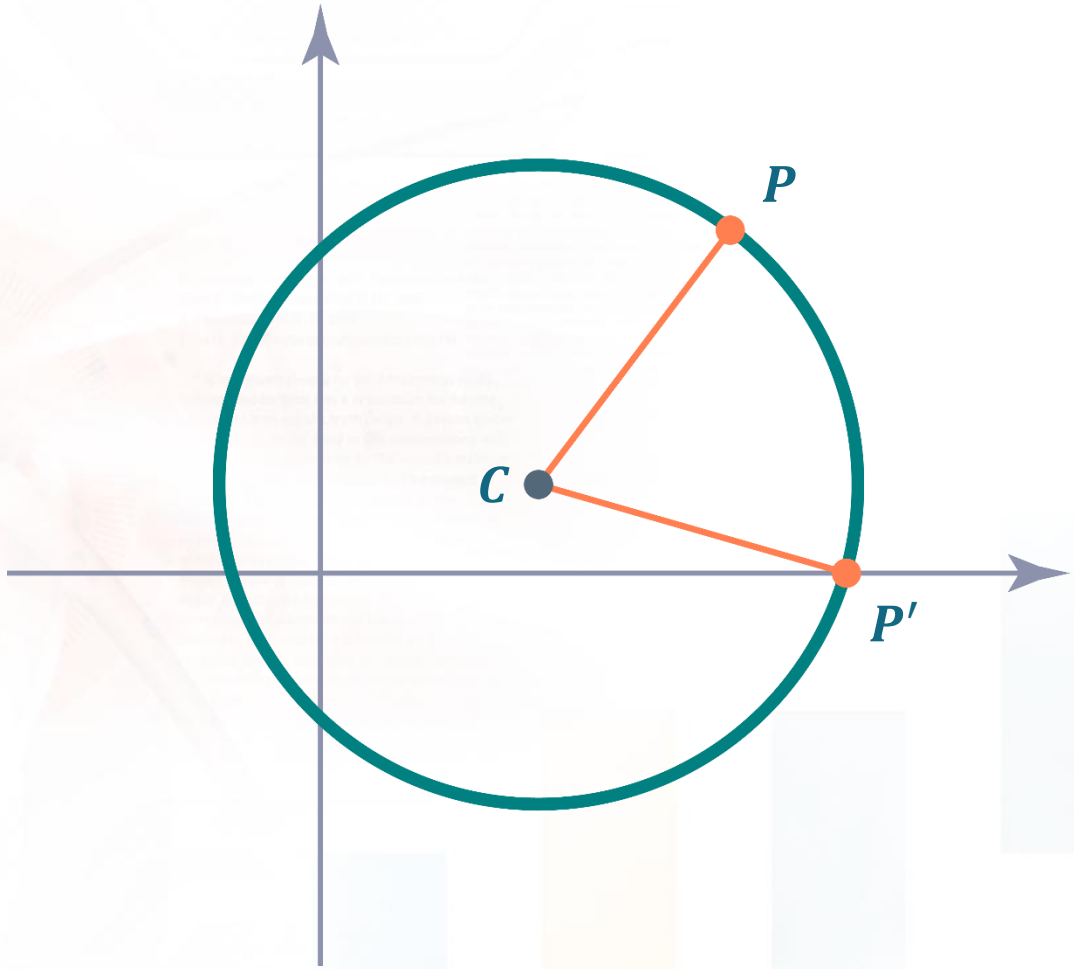
CONICAL SECTIONS



Circle Ellipse Parabola Hyperbola

DEFINITION

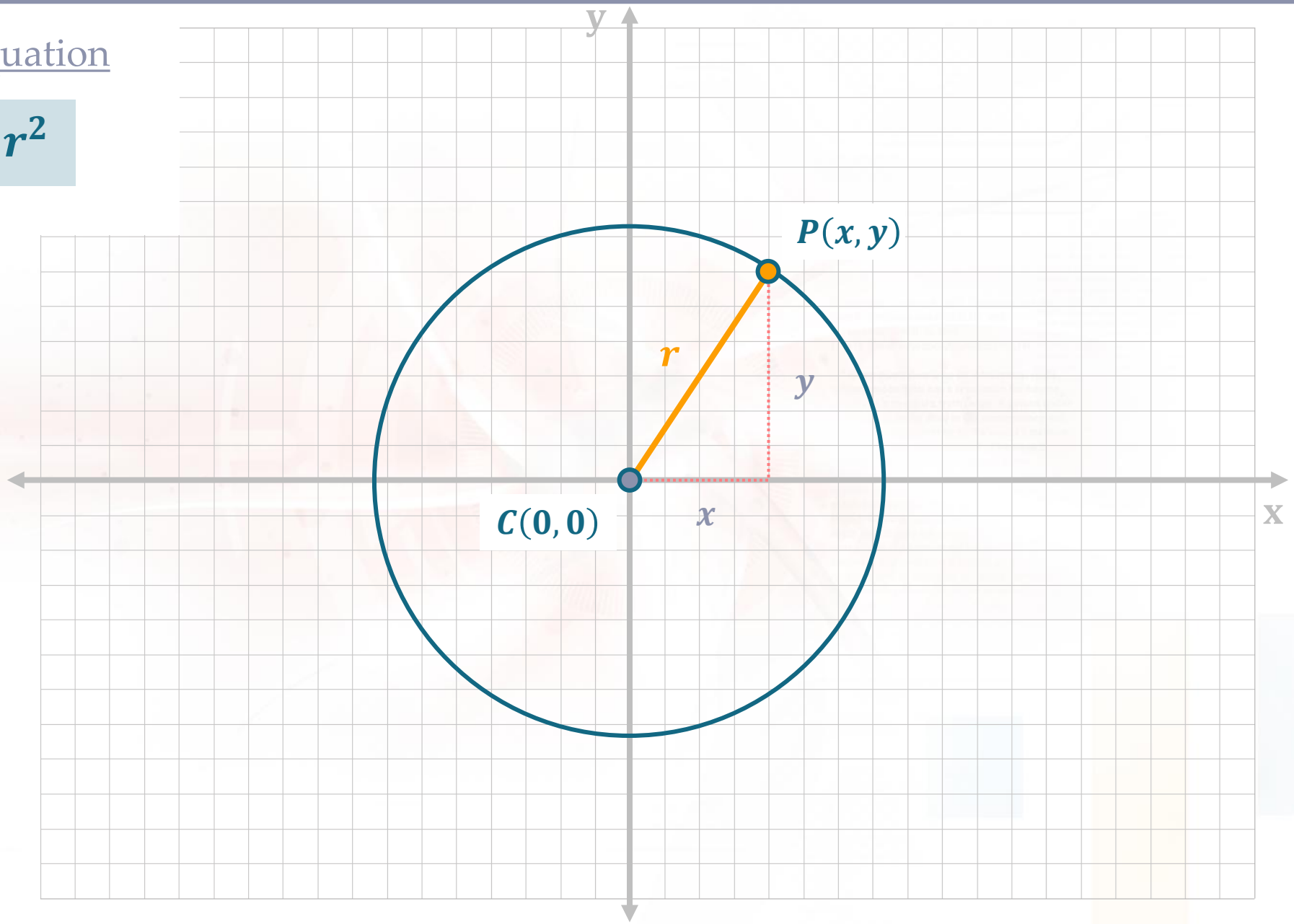
A **circle** is the set of all points in a plane that are **equidistant** (**radius**) from a fixed point (**center**)



$$d(C, P) = d(C, P')$$

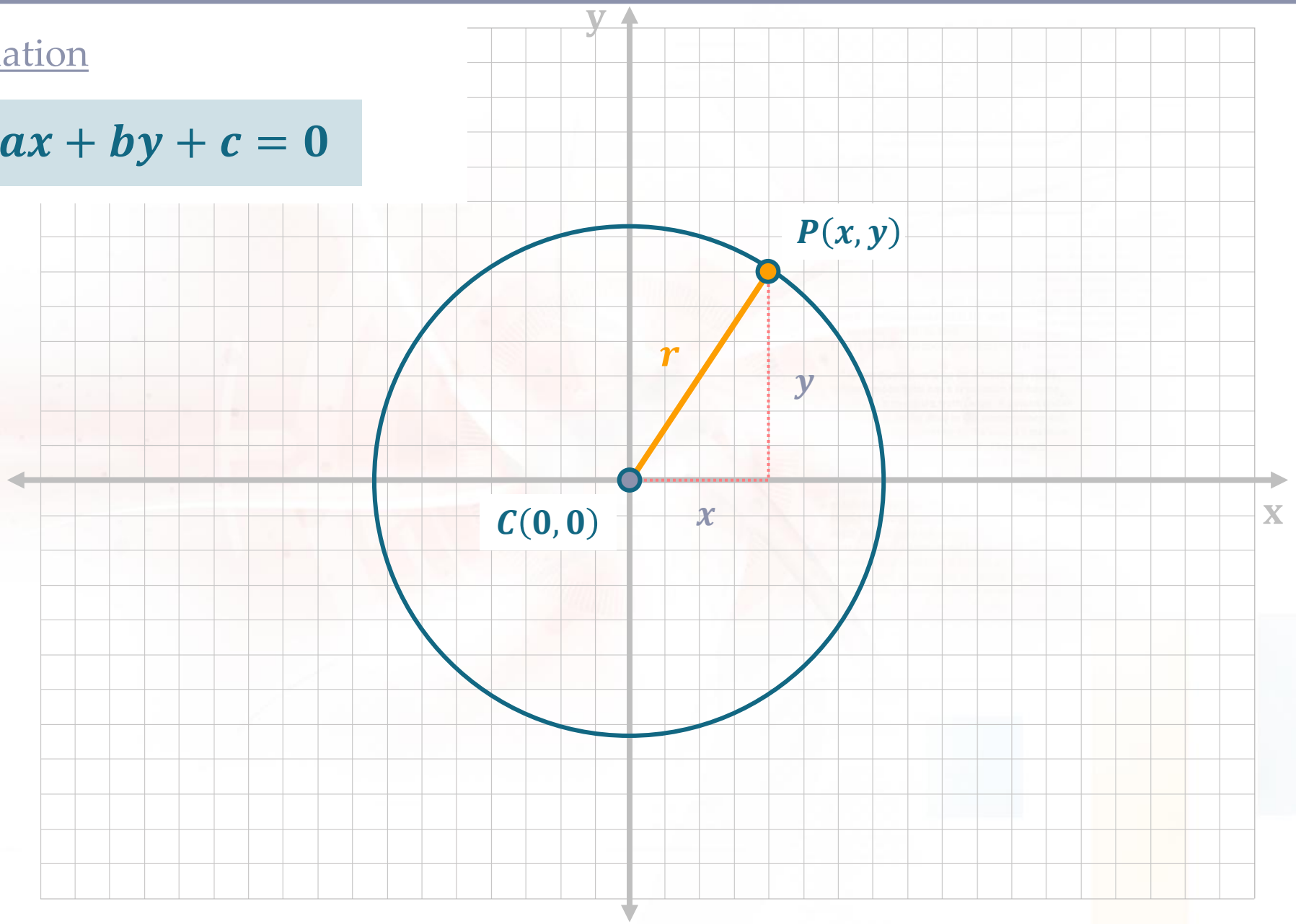
Standard Equation

$$x^2 + y^2 = r^2$$

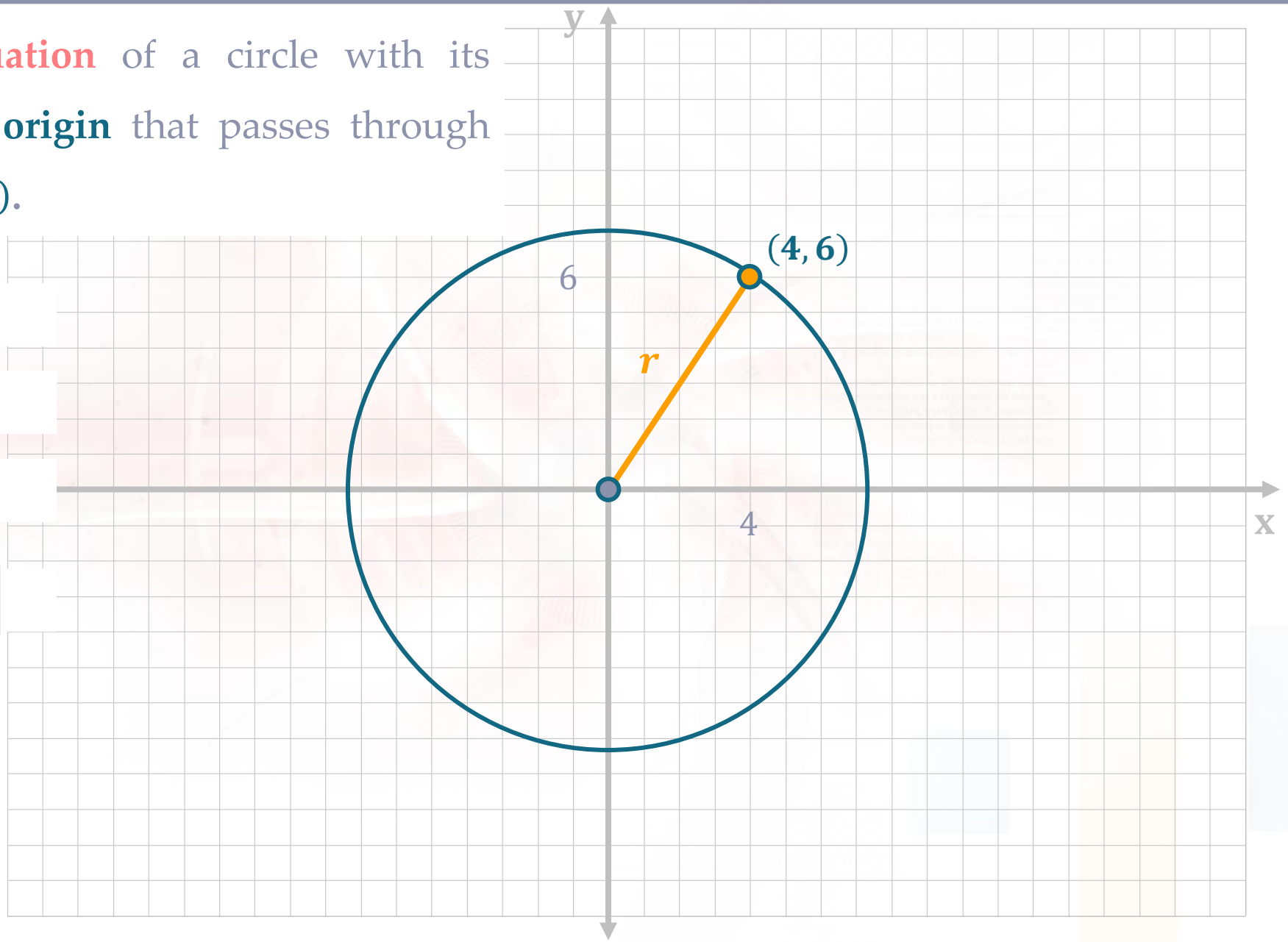


General Equation

$$x^2 + y^2 + ax + by + c = 0$$



Find the **equation** of a circle with its center at the **origin** that passes through the point $(4, 6)$.



$$x^2 + y^2 = r^2$$

$$4^2 + 6^2 = r^2$$

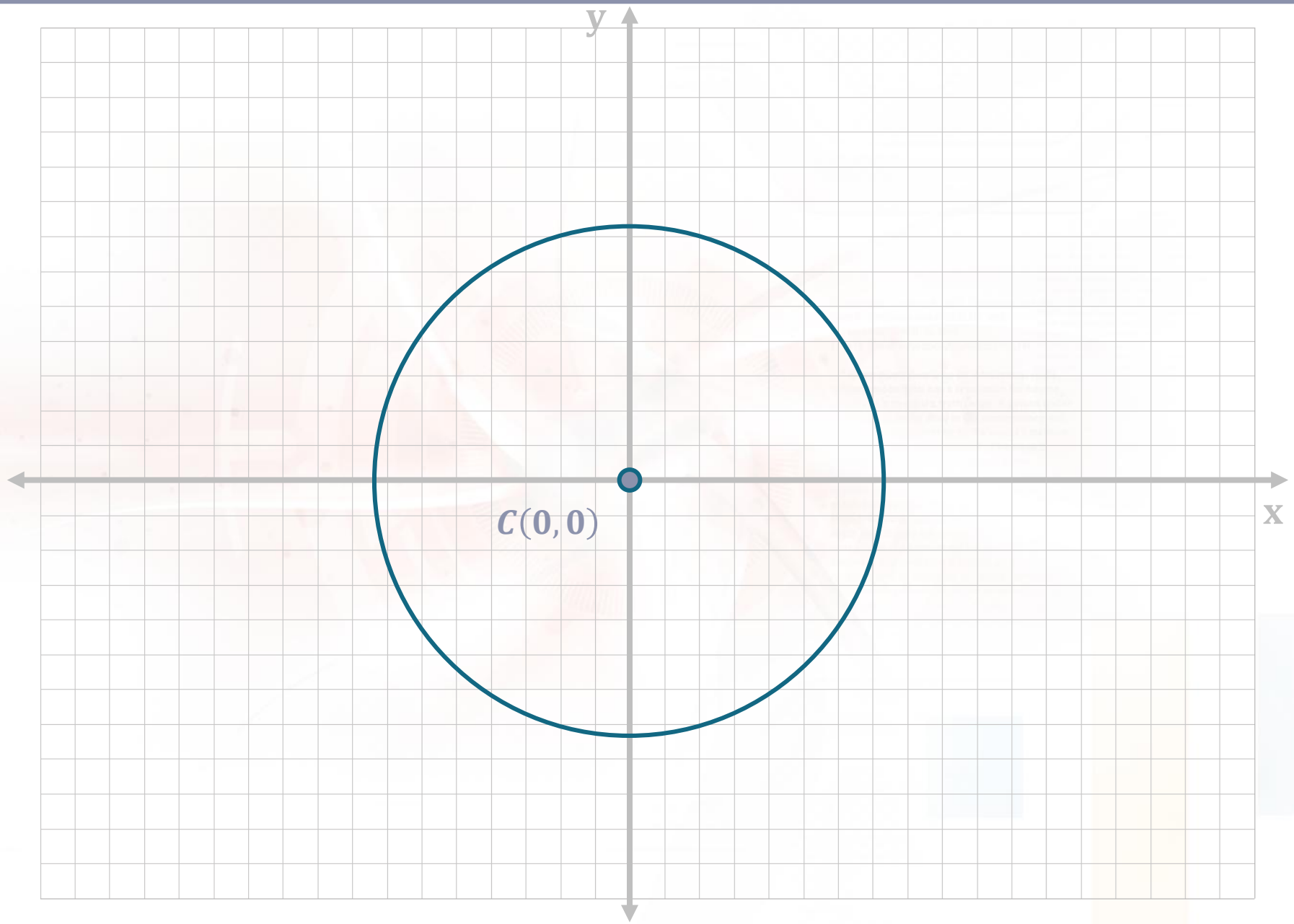
$$52 = r^2$$

$$\boxed{x^2 + y^2 = 52}$$

ans

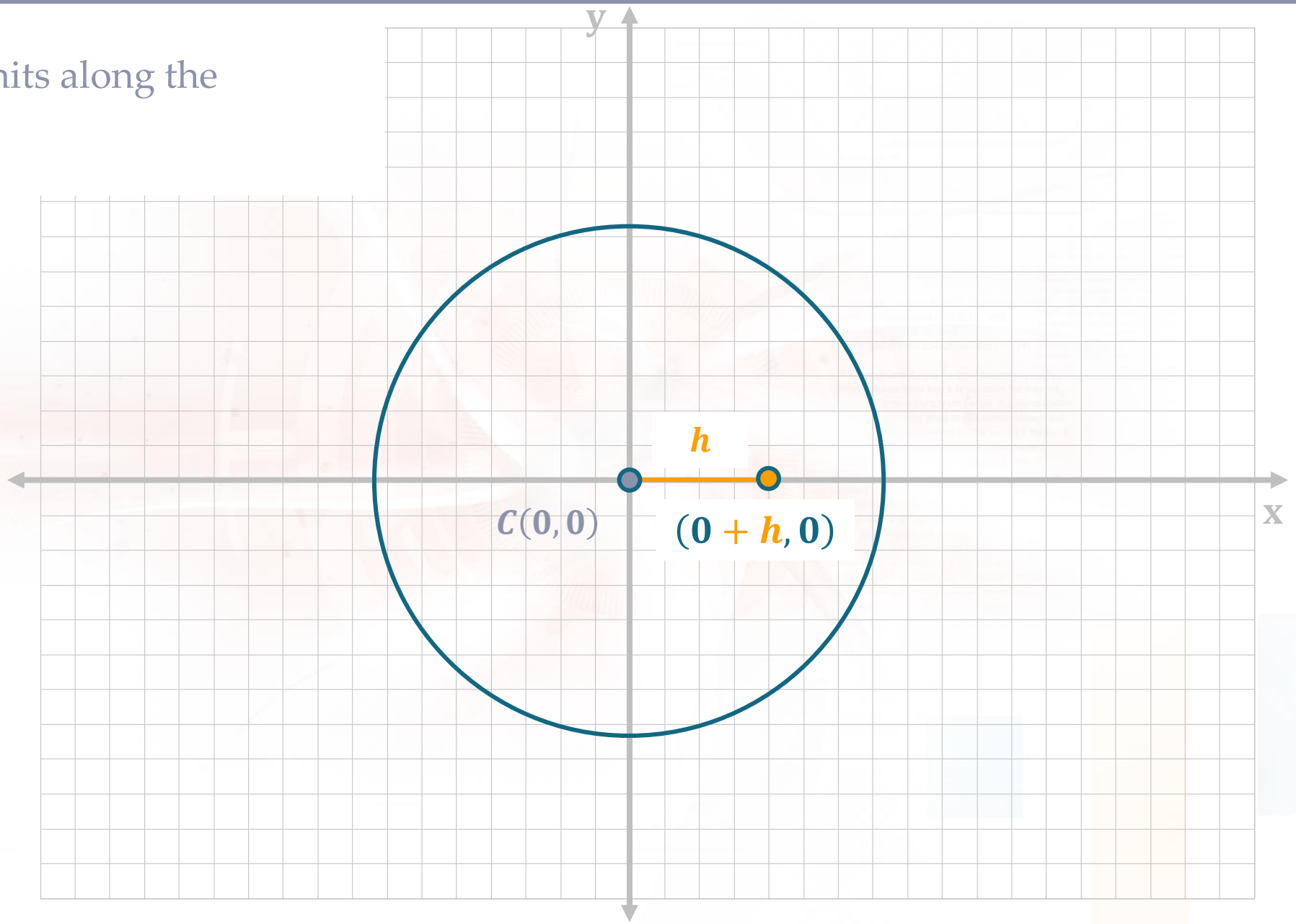
CIRCLE WITH CENTER AT $C(h, k)$

CIRCLE WITH CENTER AT THE ORIGIN



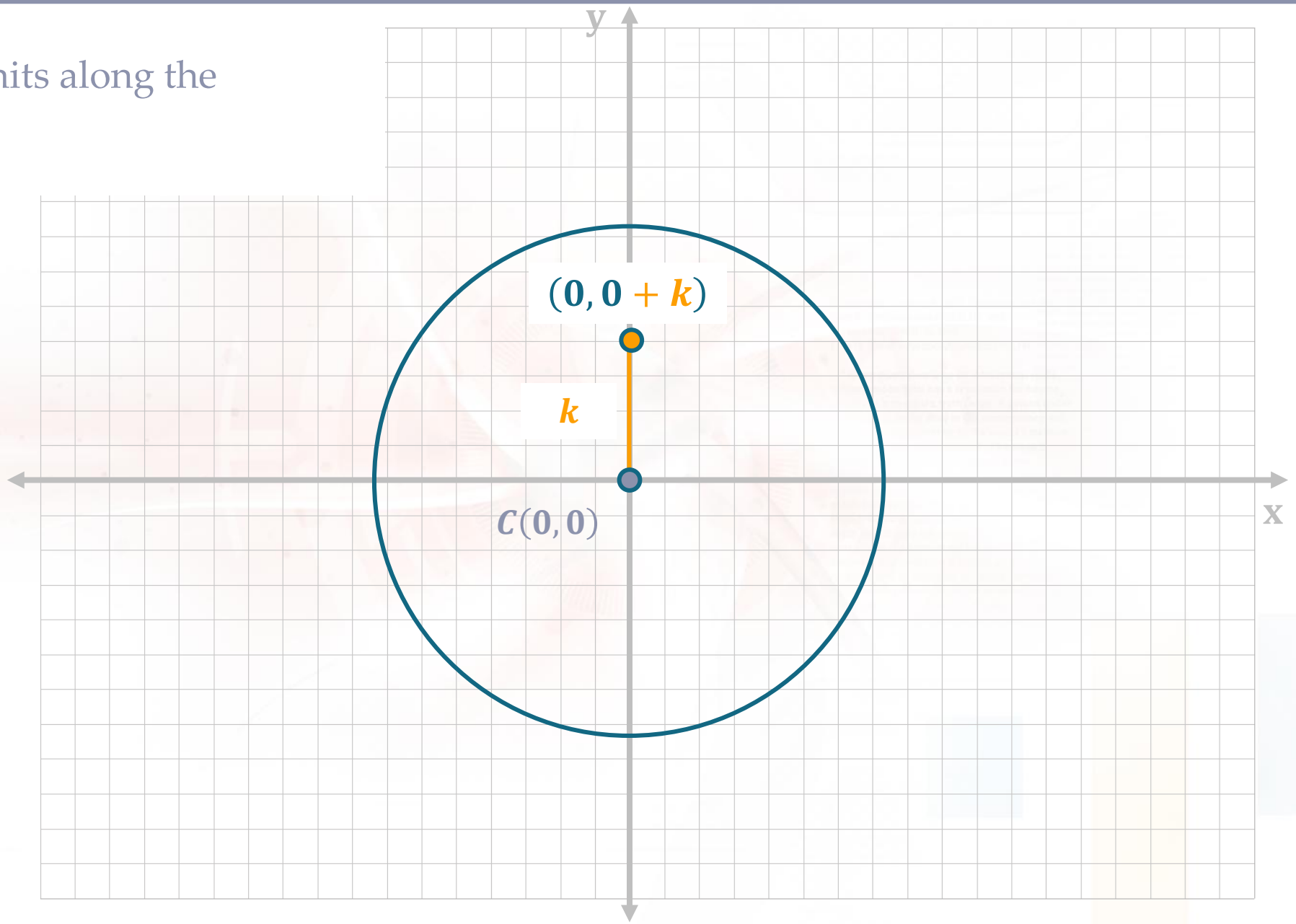
CIRCLE WITH CENTER AT $C(h, 0)$

moving h units along the x-axis



CIRCLE WITH CENTER AT $C(0, k)$

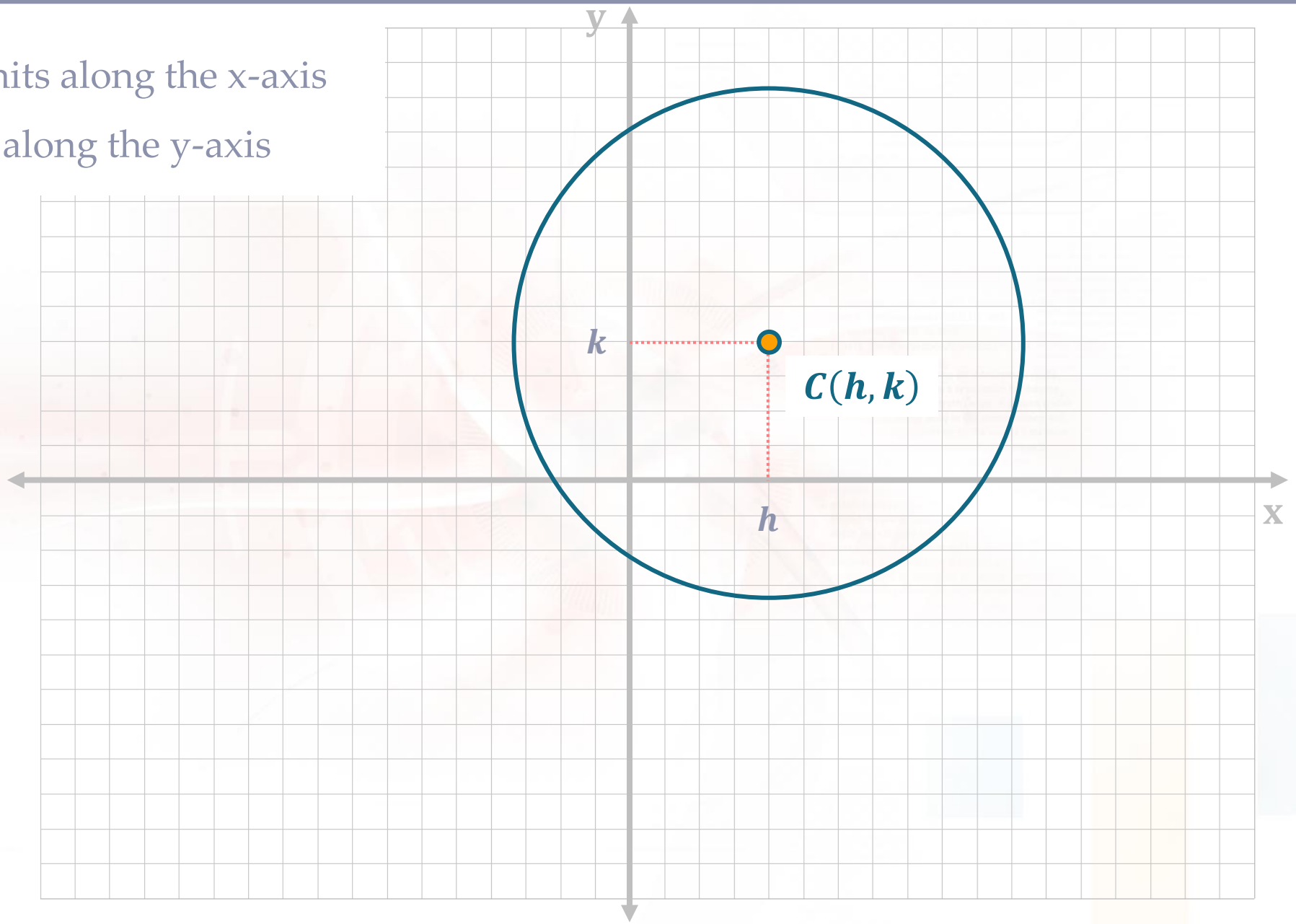
moving k units along the y-axis



CIRCLE WITH CENTER AT $C(h, k)$

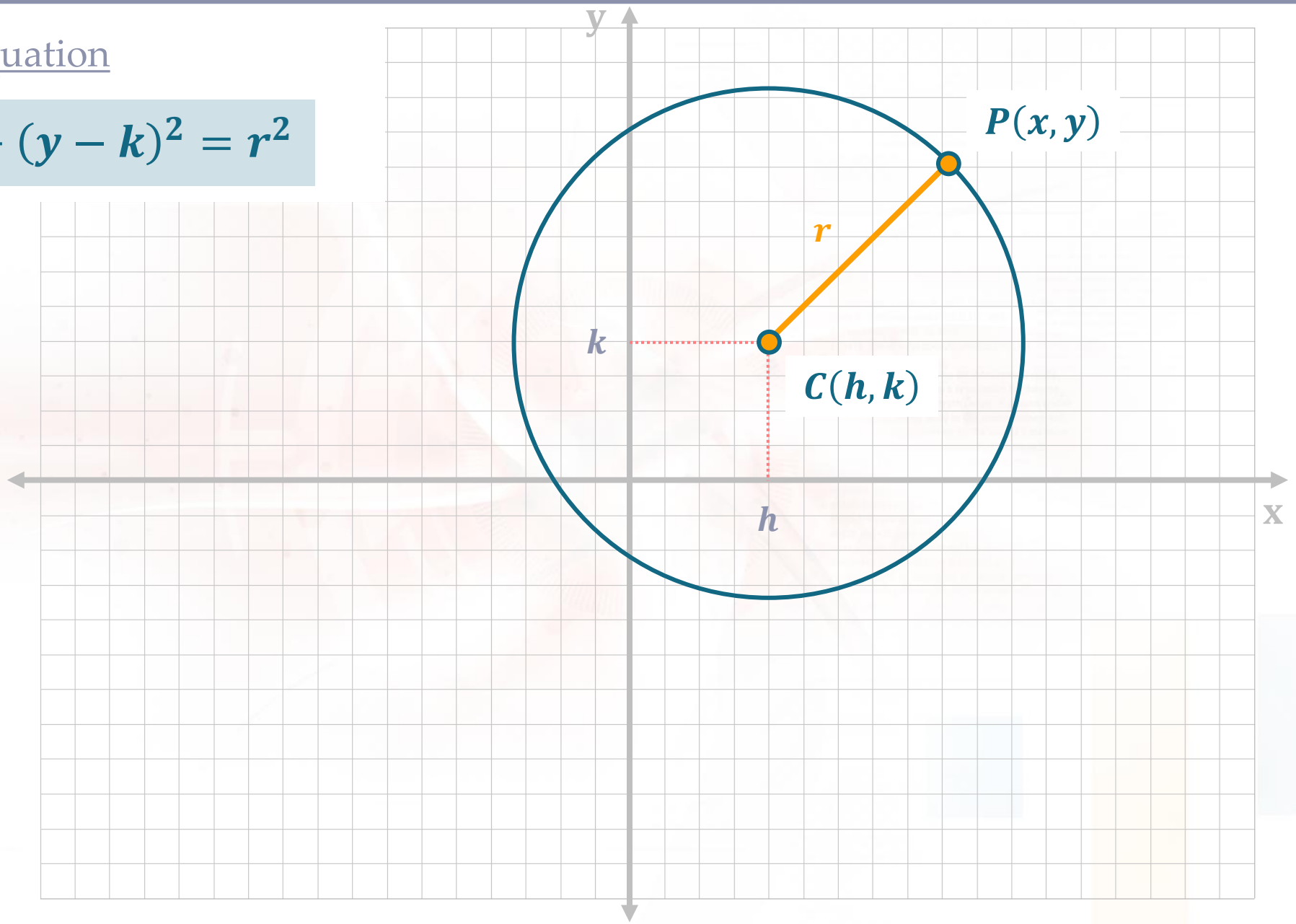
moving h units along the x-axis

and k units along the y-axis



Standard Equation

$$(x-h)^2+(y-k)^2=r^2$$



Reduce to **standard** form and **draw** the circle whose equation is

$$4x^2 + 4y^2 - 20x + 4y - 55 = 0.$$

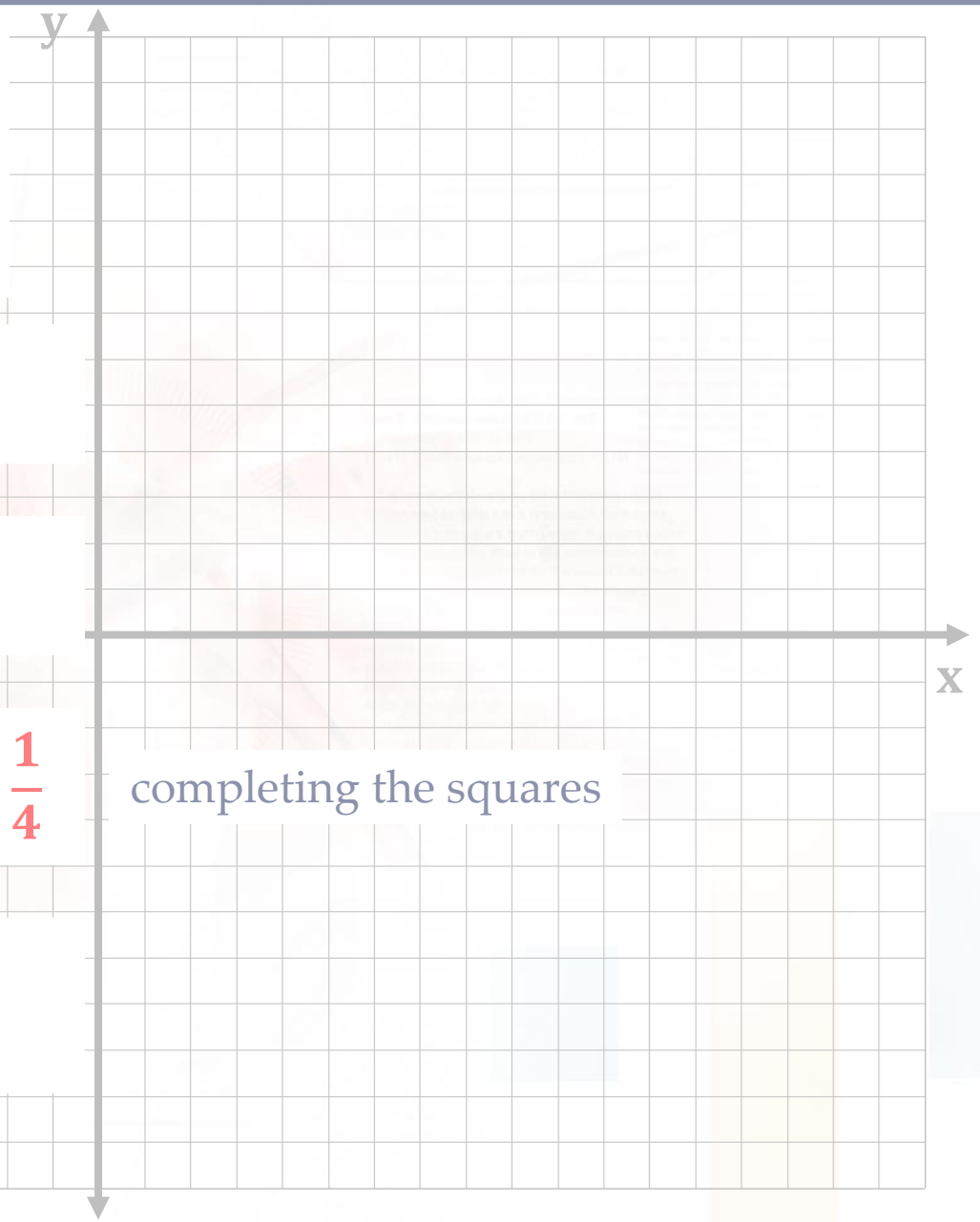
$$x^2 + y^2 - 5x + y - \frac{55}{4} = 0$$

$$(x^2 - 5x + __) + (y^2 + y + __) = \frac{55}{4}$$

$$\left(x^2 - 5x + \frac{25}{4}\right) + \left(y^2 + y + \frac{1}{4}\right) = \frac{55}{4} + \frac{25}{4} + \frac{1}{4}$$

$$\boxed{\left(x - \frac{5}{2}\right)^2 + \left(y + \frac{1}{2}\right)^2 = \frac{81}{4}}$$

ans



completing the squares

Reduce to **standard** form and **draw** the circle whose equation is

$$4x^2 + 4y^2 - 20x + 4y - 55 = 0.$$

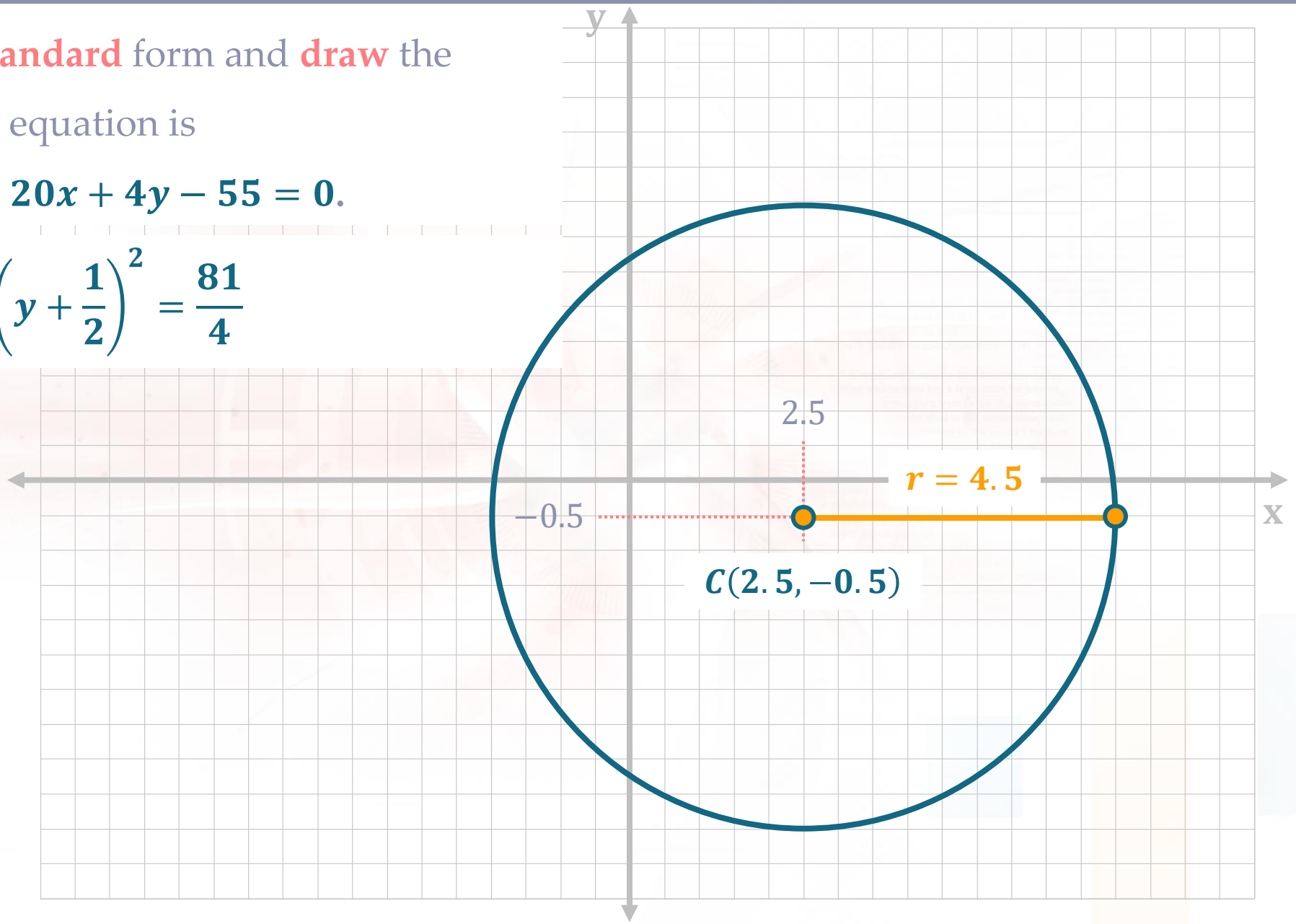
$$\left(x - \frac{5}{2}\right)^2 + \left(y + \frac{1}{2}\right)^2 = \frac{81}{4}$$

center

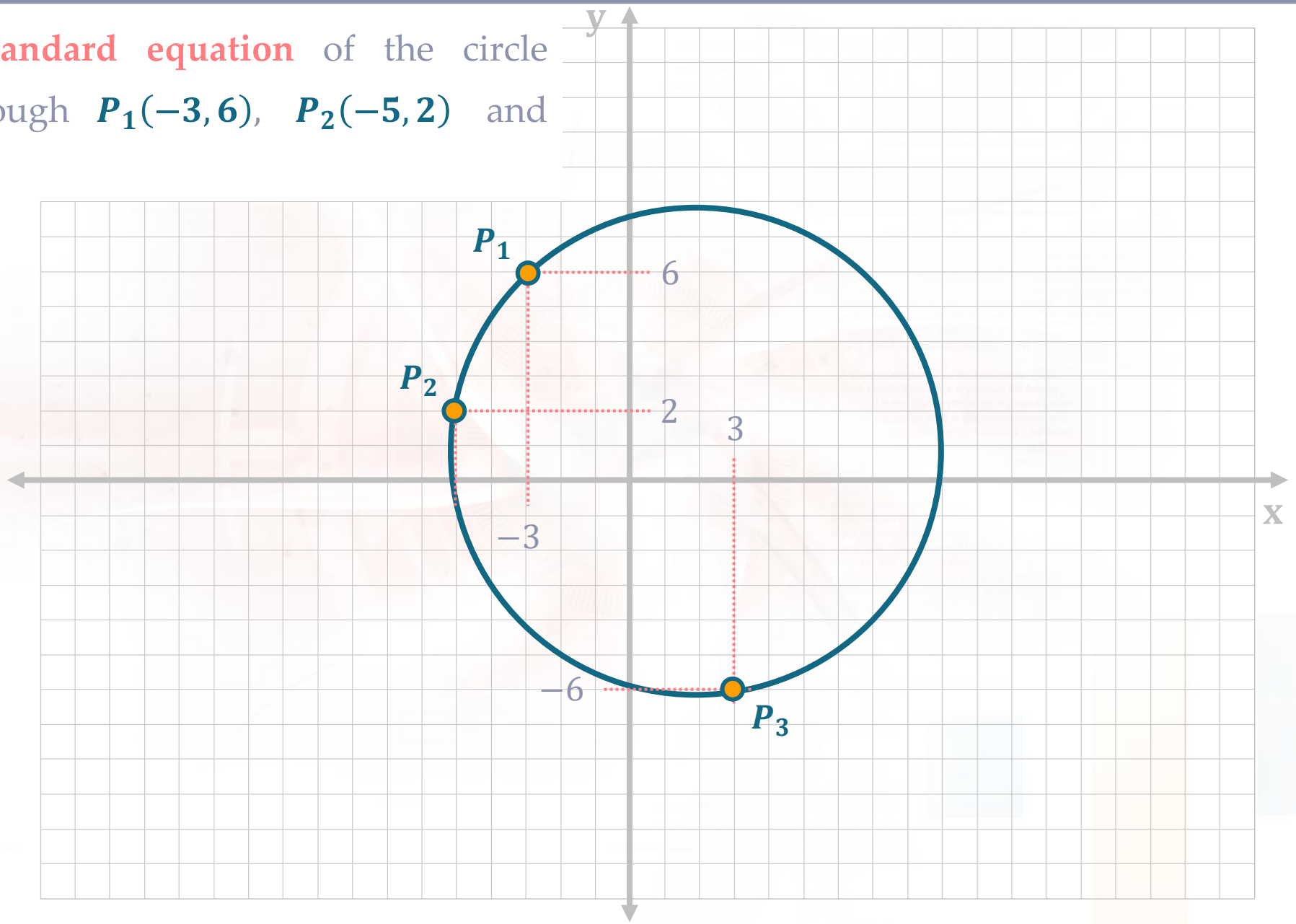
$$C\left(\frac{5}{2}, -\frac{1}{2}\right)$$

radius

$$r = \frac{9}{2}$$



Find the **standard equation** of the circle passing through $P_1(-3,6)$, $P_2(-5,2)$ and $P_3(3,-6)$.



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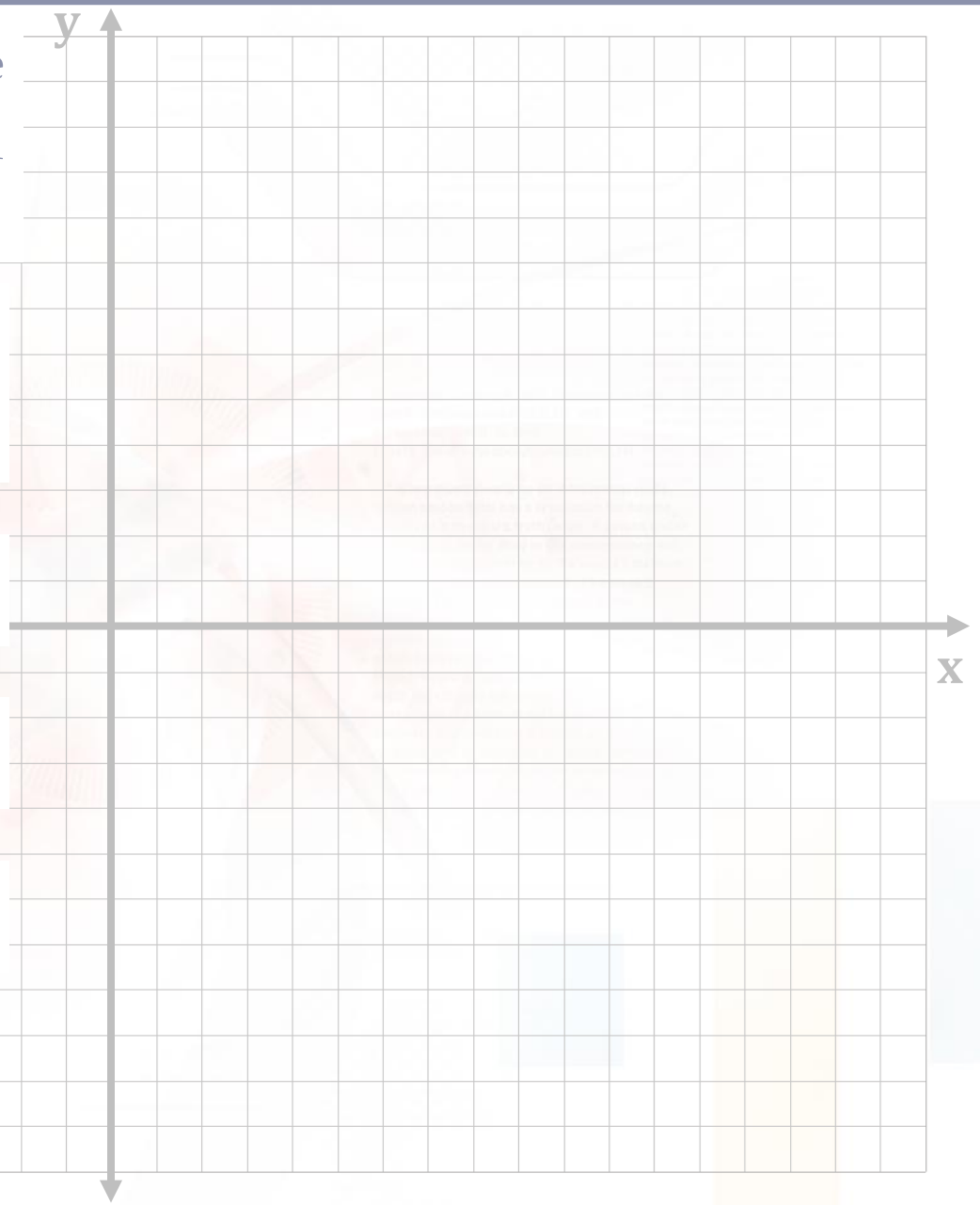


Point 1

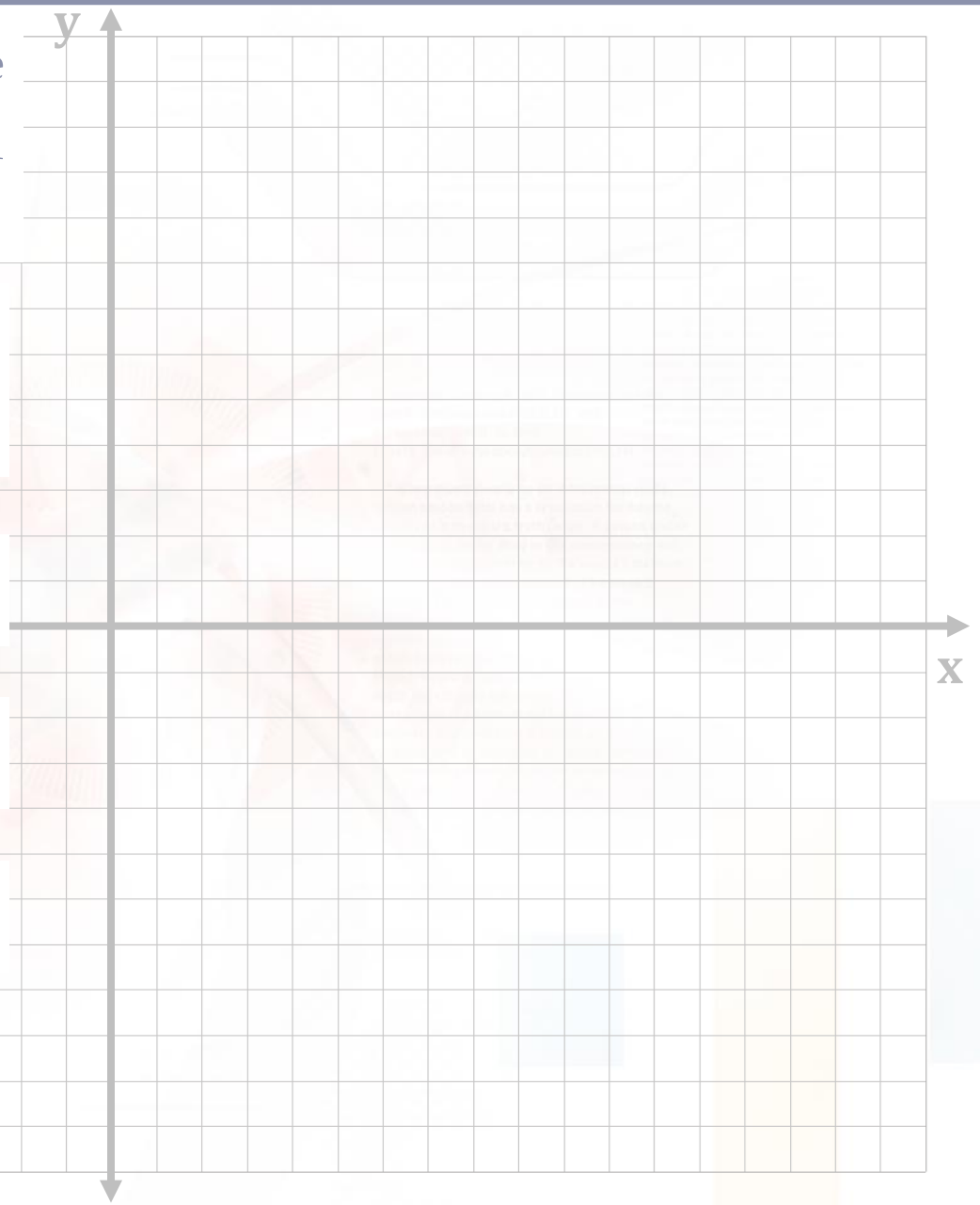
$$x^2 + y^2 + ax + by + c = 0$$

$$(-3)^2 + (6)^2 + a(-3) + b(6) + c = 0$$

$$9 + 36 - 3a + 6b + c = 0$$

$$-3a + 6b + c = -45$$ equation 1

Find the **standard equation** of the circle passing through $P_1(-3,6)$, $P_2(-5,2)$ and $P_3(3,-6)$.



Point 2

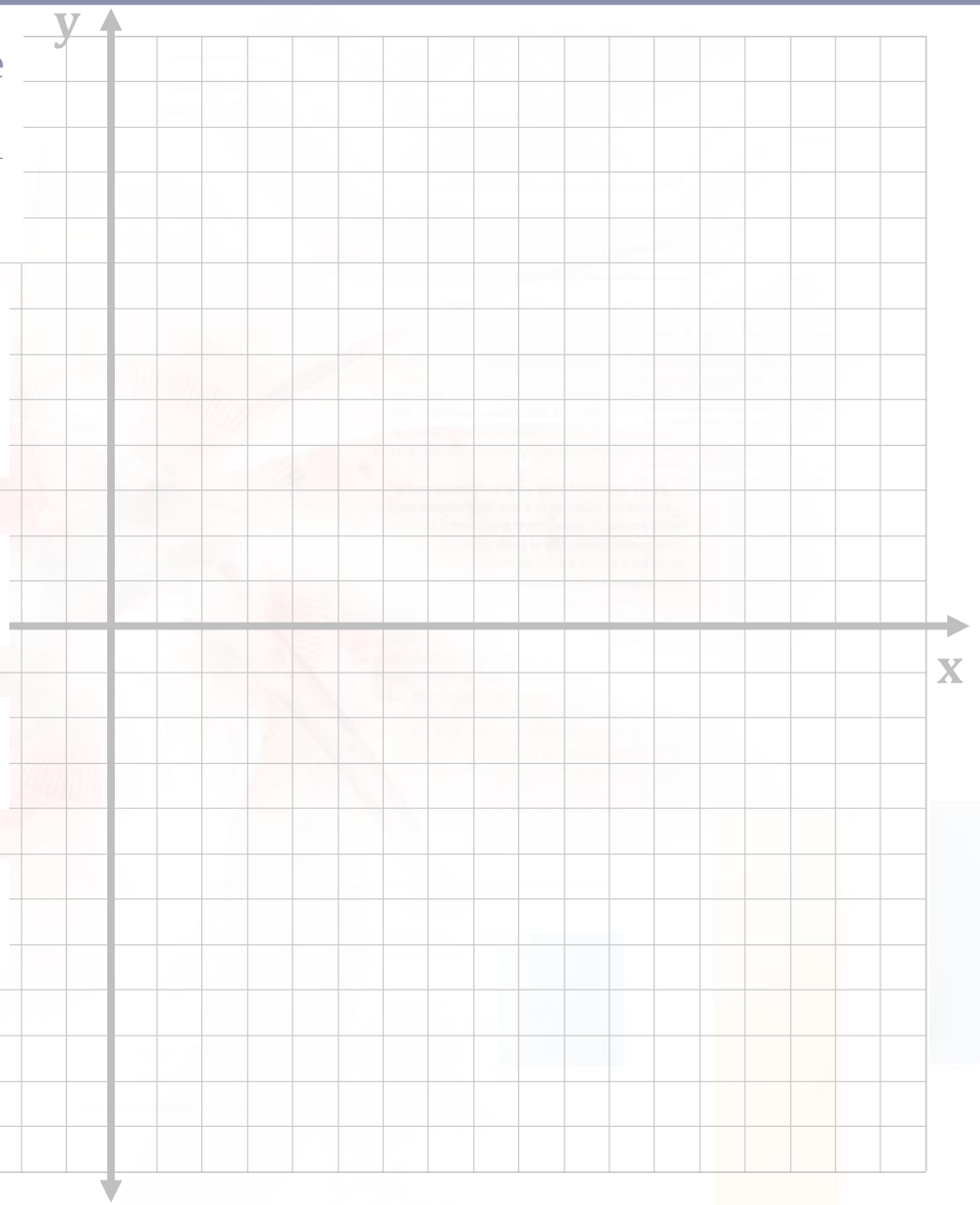
$$x^2 + y^2 + ax + by + c = 0$$

$$(-5)^2 + (2)^2 + a(-5) + b(2) + c = 0$$

$$25 + 4 - 5a + 2b + c = 0$$

$$-5a + 2b + c = -29$$ equation 2

Find the **standard equation** of the circle passing through $P_1(-3,6)$, $P_2(-5,2)$ and $P_3(3,-6)$.



Point 3

$$x^2 + y^2 + ax + by + c = 0$$

$$(3)^2 + (-6)^2 + a(3) + b(-6) + c = 0$$

$$9 + 36 + 3a - 6b + c = 0$$

$$3a - 6b + c = -45$$ equation 3

Gaussian Elimination

$$\begin{array}{cc} \begin{array}{cccc} a & b & c & \end{array} & \\ \begin{bmatrix} -3 & 6 & 1 & -45 \\ -5 & 2 & 1 & -29 \\ 3 & -6 & 1 & -45 \end{bmatrix} & \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

System of Linear Equations

$$-3a + 6b + c = -45$$

$$-5a + 2b + c = -29$$

$$3a - 6b + c = -45$$

Gaussian Elimination

$$\begin{array}{cc} \begin{array}{cccc} a & b & c & \\ -15 & 30 & 5 & -225 \end{array} & \\ \begin{bmatrix} -3 & 6 & 1 & -45 \\ -15 & 6 & 3 & -87 \\ -5 & 2 & 1 & -29 \\ 3 & -6 & 1 & -45 \end{bmatrix} & \begin{array}{l} r_1 \\ \\ r_2 \\ r_3 \end{array} \end{array} \quad \leftarrow 5r_1 - 3r_2$$

$$\begin{array}{cc} \begin{array}{cccc} a & b & c & \\ \end{array} & \\ \begin{bmatrix} -3 & 6 & 1 & -45 \\ 0 & 24 & 2 & -138 \\ 3 & -6 & 1 & -45 \end{bmatrix} & \begin{array}{l} r_1 \\ r_2 \\ r_3 \end{array} \end{array} \quad \leftarrow r_1 + r_2$$

$$\begin{array}{cc} \begin{array}{cccc} a & b & c & \\ \end{array} & \\ \begin{bmatrix} -3 & 6 & 1 & -45 \\ 0 & 24 & 2 & -138 \\ 0 & 0 & 2 & -90 \end{bmatrix} & \begin{array}{l} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

Gaussian Elimination

$$\begin{array}{cc} \begin{array}{cccc} a & b & c & \end{array} & \\ \begin{bmatrix} -3 & 6 & 1 & -45 \\ 0 & 24 & 2 & -138 \\ 0 & 0 & 2 & -90 \end{bmatrix} & \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

from equation 3

$$2c = -90$$

$$c = \frac{-90}{2}$$

$$c = -45$$

Gaussian Elimination

$$\begin{array}{cccc} a & b & c & \\ \end{array}$$
$$\begin{bmatrix} -3 & 6 & 1 & -45 \\ 0 & 24 & 2 & -138 \\ 0 & 0 & 2 & -90 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \\ r_3 \end{array}$$

from equation 2

$$24b + 2c = -138$$

$$24b + 2(-45) = -138$$

$$24b = -138 + 90$$

$$b = \frac{-48}{24}$$

$$b = -2$$

Gaussian Elimination

$$\begin{array}{cccc} a & b & c & \\ \end{array}$$
$$\begin{bmatrix} -3 & 6 & 1 & -45 \\ 0 & 24 & 2 & -138 \\ 0 & 0 & 2 & -90 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \\ r_3 \end{array}$$

from equation 2

$$-3a + 6b + c = -45$$

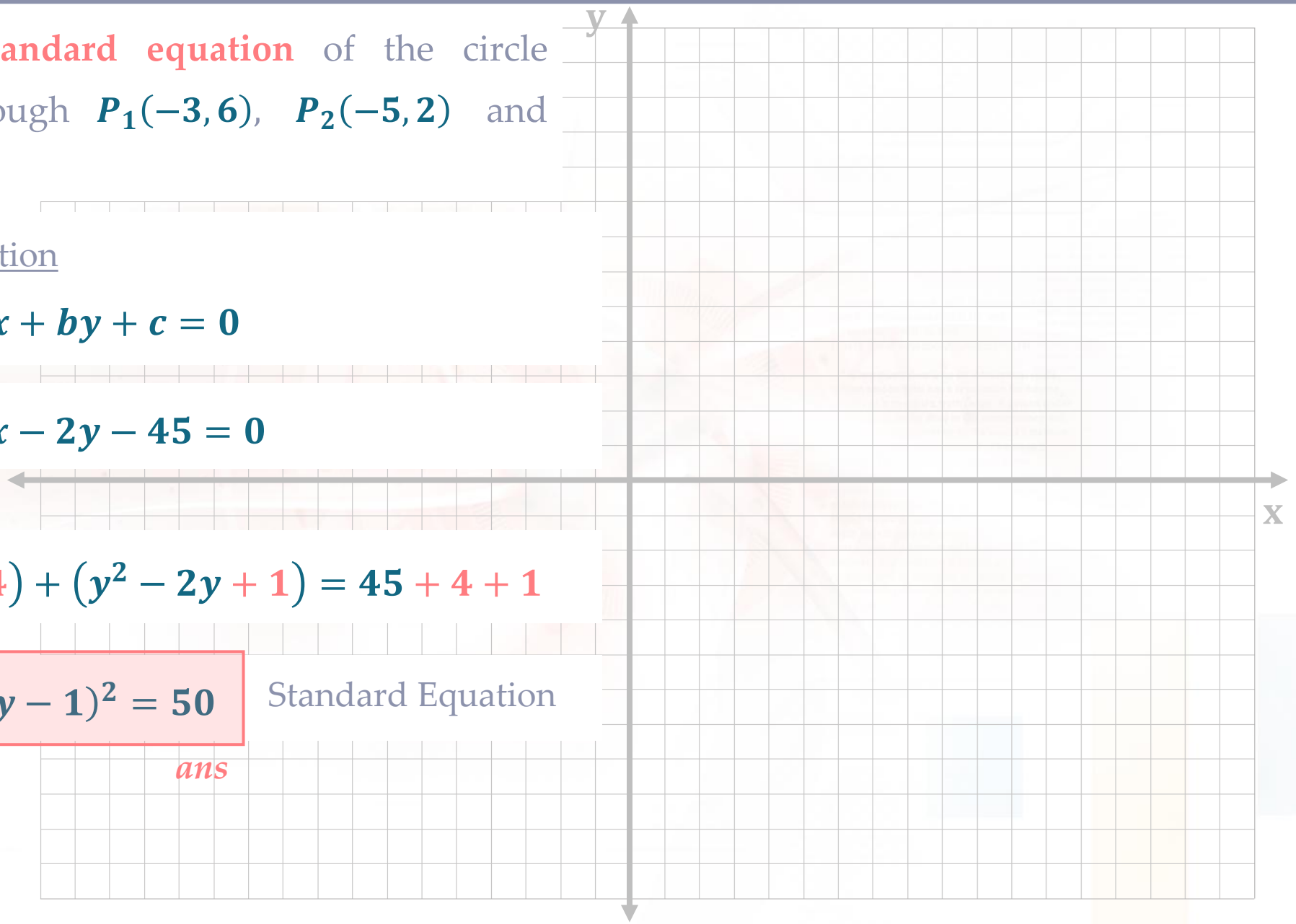
$$-3a + 6(-2) - 45 = -45$$

$$-3a = 12$$

$$a = \frac{12}{-3}$$

$$a = -4$$

Find the **standard equation** of the circle passing through $P_1(-3,6)$, $P_2(-5,2)$ and $P_3(3,-6)$.



General Equation

$$x^2 + y^2 + ax + by + c = 0$$

$$x^2 + y^2 - 4x - 2y - 45 = 0$$

$$(x^2 - 4x + 4) + (y^2 - 2y + 1) = 45 + 4 + 1$$

$$\boxed{(x-2)^2 + (y-1)^2 = 50}$$ Standard Equation

ans

Find the **standard equation** of the circle passing through $P_1(-3,6)$, $P_2(-5,2)$ and $P_3(3,-6)$.

Standard Equation

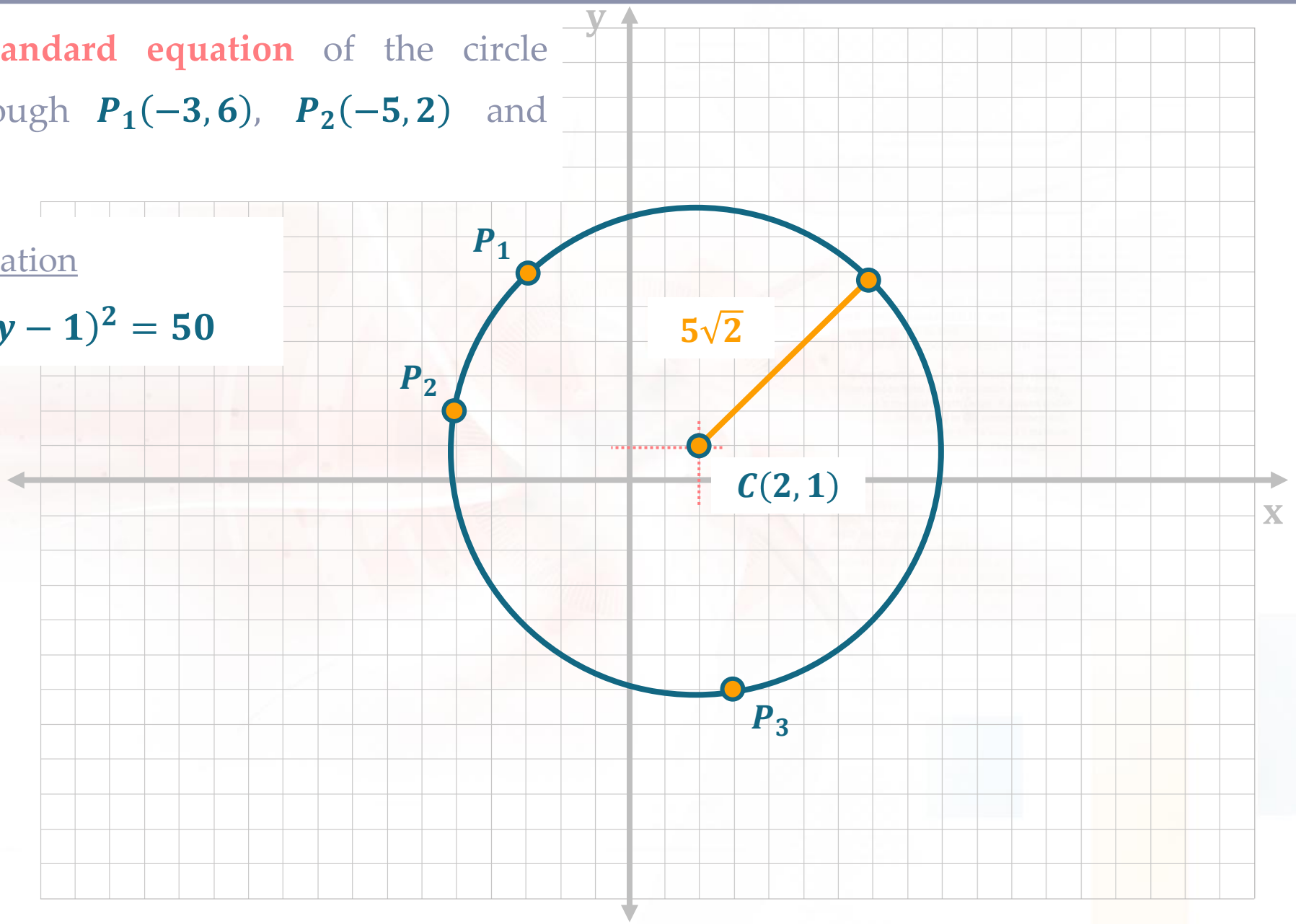
$$(x-2)^2+(y-1)^2=50$$

center

$C(2,1)$

radius

$r=5\sqrt{2}$



SEATWORK